

Notes: Boutros (JFE, 2026)
Windfall Income Shocks with Finite Planning Horizons

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1 Summary

This paper studies why households, including households with high income and liquid wealth, often spend a large share of one-time windfall income shocks instead of smoothing them over the rest of life. The central idea is **bounded intertemporal rationality**: households are able to optimize, but forming and implementing a new intertemporal plan is cognitively costly. After a windfall income shock, the household chooses a finite planning horizon over which to reoptimize, pays a planning cost that rises with the length of that horizon, and then returns to the original long-run plan.

The model combines two behavioral ingredients. First, windfall income is placed in a separate mental account, so it is not fully fungible with regular income and regular wealth at the moment of receipt. Second, reoptimization is costly, so households choose how far into the future to revise their consumption-savings plan. These assumptions imply that small shocks are optimally handled with short, local planning horizons, generating high immediate marginal propensities to consume (MPCs). Large shocks make the benefits of smoothing more important, leading households to choose longer horizons and lower MPCs.

The model delivers two important monotonicity results. The optimal planning horizon increases with the size of the shock, and it also increases with household income and wealth. Intuitively, larger shocks create larger gains from smoothing. Wealthier households also benefit more from spreading extra consumption over additional future periods because their marginal utility falls less steeply when they move consumption into later dates.

The quantitative part calibrates planning costs using the 2008 Economic Stimulus Payments. The model matches three facts that standard one- and two-asset models have difficulty matching at the same time: unconstrained households can still have large consumption responses; the fraction of households with a positive response rises with the size of the shock; and conditional on responding, larger shocks have lower MPCs. The calibrated model then becomes a laboratory for fiscal stimulus design and for the welfare effects of policies that reduce planning costs.

2 Big picture

- **Question.** How do the cognitive costs of financial planning shape household consumption responses to windfall income shocks?
- **Main answer.** Households choose finite planning horizons. They smooth a shock only over the horizon for which the benefits of smoothing justify the cognitive cost of reoptimization.
- **Mechanism.** Small shocks trigger local thinking and short horizons; large shocks trigger more global thinking and longer horizons.
- **Key model object.** The planning cost function $\Phi(k)$, where k is the length of the reoptimization horizon.
- **Key predictions.** Optimal k is increasing in shock size, income, and wealth.
- **Empirical target.** The 2008 Economic Stimulus Payments provide variation in shock size relative to monthly income and liquid wealth.
- **Policy implication.** Stimulus programs that target smaller payments to poorer households can generate larger aggregate spending responses; financial literacy that lowers planning costs may create larger welfare gains for wealthier households.

3 Incremental contribution of the paper

- **A new bounded-rationality friction.** Instead of modeling households as simply hand-to-mouth, inattentive, or constrained, the paper models households as rational over a chosen finite planning horizon.
- **Unifies intensive and extensive margins.** The model explains why more households respond when shocks are larger, while the MPC among responders can decline with shock size.
- **Applies mental accounting to dynamic planning.** Windfalls enter a separate account, and households decide how many future periods to revise before returning to their existing lifetime plan.
- **Quantitative discipline.** Planning costs are calibrated to consumption responses to the 2008 Economic Stimulus Payments using GMM targets.
- **Policy laboratory.** The calibrated model can compare the aggregate MPC effects of alternative stimulus designs and the welfare effects of lower planning costs.

4 Motivating evidence and model intuition

4.1 Why finite horizons?

The paper starts from the observation that many consumption responses to windfall income shocks are concentrated in a short window after receipt. Under full rationality, an unconstrained household should reoptimize its lifetime plan after every shock. In contrast, the paper argues that households face cognitive costs in making and implementing new intertemporal plans, so they choose how much of the future to reconsider.

The central simplifying assumption is that, for windfall shocks, households choose a planning horizon k . During the next k periods, they form new consumption and savings plans. After that horizon is completed, they return to the previously formed long-term plan.

4.2 Three-period intuition

Figures 1 and 2 give the intuition. Without planning costs, a financially unconstrained household spreads an income shock evenly over remaining periods. With planning costs, the household compares the benefit of smoothing into extra future periods with the incremental cost of making a longer plan. Short planning horizons therefore generate front-loaded spending, even for households that are not borrowing constrained.

A key distinction is between constrained and unconstrained households:

- **Constrained households** have high MPCs because they already want to borrow against future income but cannot.
- **Unconstrained households** can also have high MPCs because they choose not to incur the planning cost required to smooth a relatively small windfall over many future periods.

5 Baseline long-term plan

The outer layer of the model is a standard life-cycle consumption-savings problem. The household has wealth w , income y , consumption c , savings s , and leisure ℓ . It maximizes expected lifetime

utility subject to a budget constraint and an exogenous borrowing limit:

$$V_t(w, y) = \max_{c_t, s_t} u(c_t, \ell) + \beta E_t V_{t+1}(w', y'),$$

$$c_t + R^{-1} s_t = y_t + w_t, \quad s_t \geq -\underline{b},$$

$$w' = s_t(w, y) = R(y_t + w_t - c_t(w, y)).$$

This layer represents the long-term plan that the household has already formed before the windfall shock arrives. If no windfall occurs, the household follows this plan as in a standard consumption model.

6 Windfall income shocks and bounded intertemporal rationality

6.1 Mental accounts

When a windfall shock Δ arrives, the household does not simply add it to regular wealth. Instead, the shock enters a separate windfall account. This is what makes the shock “special” relative to ordinary income and wealth. The household then decides how many periods to spend down this windfall account.

6.2 Reoptimization problem

For a chosen horizon k , the household forms new consumption and savings plans over periods t to $t + k - 1$. Planning costs are paid at the start of the reoptimization window and depend only on k :

$$\Phi(k), \quad \Phi(0) = 0, \quad \Phi(k) \text{ increasing in } k.$$

A compact representation of the problem is

$$\max_{k, \{c_\tau, s_\tau\}_{\tau=t}^{t+k-1}} E_t \left[u(c_t, \ell - \Phi(k)) + \sum_{\tau=t+1}^{t+k-1} \beta^{\tau-t} u(c_\tau, \ell) + \sum_{q=t+k}^T \beta^{q-t} u(c_q, \ell) \right],$$

subject to the periodic budget constraints and the borrowing constraint. The first period bears the planning cost, the intermediate periods follow the temporary windfall plan, and the post-horizon periods return to the original lifetime plan.

6.3 Optimal planning horizon

The model gives two core monotonicity results:

$$\begin{aligned} \Delta' > \Delta &\Rightarrow k_t^*(\Delta'; w, y) \geq k_t^*(\Delta; w, y), \\ w' > w \text{ or } y' > y &\Rightarrow k_t^*(\Delta; w', y') \geq k_t^*(\Delta; w, y). \end{aligned}$$

The first result says that larger shocks justify longer planning horizons. The second result says that richer households choose longer horizons for a given shock. This second prediction is important because it separates the paper from a pure borrowing-constraint model: wealth affects the planning horizon even when households are not constrained.

7 Calibration and empirical strategy

7.1 External calibration

The model is calibrated in two steps. First, the outer life-cycle model is calibrated using standard estimates from the literature and data sources. Income dynamics are based on estimates from Gelman (2021). Preference parameters follow Kaplan and Violante (2014). Financial-market parameters use the Federal Reserve Board and the Survey of Consumer Finances.

Table 2 summarizes the long-term parameters. Figure 3 shows that the calibrated long-term model matches the central part of the liquid-wealth-to-monthly-income distribution reasonably well, although it cannot match the upper tail of very liquid households.

7.2 Internal calibration using stimulus payments

The planning cost function $\Phi(k)$ is calibrated to match consumption responses to the 2008 Economic Stimulus Payments. The empirical design uses the fact that stimulus payments differed in size relative to households' monthly income and liquid wealth. Households are sorted into terciles of payment size relative to monthly income.

The empirical stimulus-response regression follows the structure

$$\Delta c_{it} = \alpha + \beta \cdot ESP_{it} + \delta z_{it} + \gamma_t + u_{it},$$

and the paper also estimates interactions between ESP_{it} and the relative-payment terciles:

$$\Delta c_{it} = \alpha + \beta_1 ESP_{it} + \sum_{j=2}^3 \beta_j ESP_{it} \times \mathbf{1}\{\text{Tercile } j\}_{it} + \delta z_{it} + \gamma_t + u_{it}.$$

The main object is the implied three-month MPC. Table 4 reports the empirical spending response. Table 5 shows that the calibrated model matches the six GMM targets closely and also performs external validation against evidence from Gelman (2021).

7.3 Calibrated planning costs

Figure 4 shows the calibrated planning cost function. The cost rises with horizon length and is economically meaningful but not implausibly large: the paper interprets it as the time and effort cost of forming new financial plans. The key role of $\Phi(k)$ is not to make households irrational, but to make lifetime reoptimization costly enough that finite horizons become optimal.

8 Main quantitative results

8.1 MPC and liquid wealth

Figure 5 compares empirical evidence and model predictions. The empirical pattern is that MPCs decline with liquid wealth but remain positive even for relatively liquid households. Standard one-asset and two-asset models produce too little spending by liquid households because unconstrained households smooth the shock over their whole lifetime. The bounded intertemporal rationality model produces larger MPCs for liquid households because they optimally choose shorter planning horizons for smaller shocks.

8.2 Shock size and extensive margin

Figure 6 focuses on the fraction of households with a positive MPC. Empirically, the fraction of respondents with a positive consumption response increases with shock size. The BIR model can generate this positive extensive-margin effect because larger shocks make it worthwhile for more households to incur the planning cost and reoptimize.

8.3 Shock size and intensive margin

Figure 7 focuses on the MPC conditional on responding. Empirically, larger shocks generate lower MPCs. The BIR model matches this because larger shocks justify longer planning horizons and therefore greater smoothing. Standard models have difficulty generating both the positive extensive-margin effect and the negative intensive-margin effect.

9 Policy implications

9.1 Stimulus design

The model has direct implications for stimulus policy. Since aggregate consumption responses depend on shock size relative to income and wealth, stimulus design can change the aggregate MPC. Table 6 compares the observed 2008 benchmark with alternative cost-neutral redistributions. Moving payments toward the lowest-income households increases the aggregate MPC substantially because these households choose shorter planning horizons and spend more quickly.

9.2 Financial literacy and planning costs

The paper also studies a policy that lowers planning costs, interpreted as an improvement in financial literacy or access to financial advice. Figure 8 translates reductions in planning costs into equivalent windfall shocks, and Figure 9 shows the distribution of welfare gains. The striking implication is that lowering planning costs can benefit wealthier households more, because wealthier households lose more from being unable to smooth over longer horizons.

10 Interpretation and caveats

- **Bounded rationality is horizon-specific.** Households are not modeled as globally irrational. They optimize within the horizon that they choose to reconsider.
- **Planning costs are reduced-form.** The model does not separately identify attention costs, cognitive costs, adjustment costs, or the cost of using financial advice.
- **Mental accounting is central.** Without treating windfalls as separate accounts, the model would collapse toward standard consumption smoothing behavior.
- **The 2008 stimulus is a specific context.** The calibration uses a particular U.S. policy episode; other windfalls may differ in salience, timing, or perceived permanence.
- **Policy effects are distributional.** Stimulus design affects aggregate spending, but changes in planning costs affect welfare differently across the wealth distribution.

11 Notes-created summary tables

11.1 Research design at a glance

Element	Implementation	Purpose
Question	Consumption response to unanticipated windfall income shocks.	Explains why even liquid households can have high MPCs.
Friction	Planning cost $\Phi(k)$ for reoptimizing over k future periods.	Makes finite-horizon planning optimal.
Behavioral structure	Windfalls are mentally accounted for separately from regular wealth.	Makes windfall shocks non-fungible at the moment of receipt.
Key monotonicity	k^* rises with shock size, wealth, and income.	Generates both extensive-margin and intensive-margin shock-size effects.
Calibration	2008 Economic Stimulus Payments and GMM targets.	Disciplines the planning cost function quantitatively.
Policy counterfactuals	Alternative stimulus distributions and lower planning costs.	Studies aggregate MPC and welfare implications.

11.2 Mechanism map

Shock or household feature	Effect on planning horizon	Effect on consumption response
Small windfall	Short horizon because smoothing benefits are small relative to planning costs.	High immediate MPC.
Large windfall	Longer horizon because smoothing benefits are large.	Lower MPC conditional on responding.
Low liquid wealth / borrowing constraint	Household would like to move consumption forward even absent planning costs.	High MPC through standard constraint channel.
High liquid wealth	Household can smooth, but may not pay the cost for a long horizon.	Positive MPC through bounded-planning channel.
Lower planning costs	Longer horizons become cheaper.	More smoothing and lower MPC for affected households.

11.3 Empirical facts matched by the model

Fact	Why standard models struggle	BIR explanation
Liquid households have high MPCs	Unconstrained households should smooth shocks over life.	They choose short horizons because planning is costly.
Positive-response share rises with shock size	Standard models often predict positive responses for all households or mainly constrained households.	Larger shocks make reoptimization worthwhile for more households.
MPC falls with shock size conditional on response	Concave consumption functions alone do not jointly match extensive and intensive margins.	Larger shocks induce longer horizons and more smoothing.
Stimulus policy has distributional MPC effects	The same dollar shock has different relative size across households.	Relative shock size changes the chosen planning horizon.

12 Original figures and tables

12.1 Figure 1 and Table 1

Read: Figure 1 illustrates the three-period consumption-smoothing logic for constrained and unconstrained households. Table 1 reports the MPC path implied by different finite planning horizons in the stylized example.

Economic message: A finite horizon makes even an unconstrained household spend more immediately, because smoothing into more periods requires paying higher planning costs.

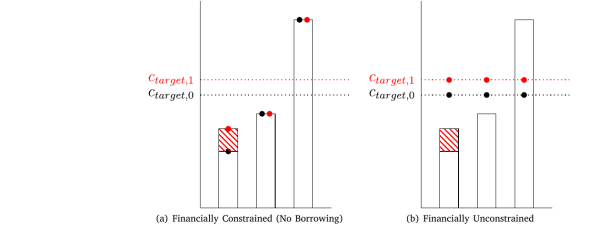


Fig. 1. Consumption smoothing in stylized model.
Note: Illustrative three-period model of consumption smoothing. Vertical bars depict income, dashed lines represent consumption target actual consumption. Black pattern illustrates initial household behavior and red pattern illustrates new behavior after $\epsilon > 0$ income shock

(a) Figure 1. Consumption smoothing in stylized model.

Table 1
MPCs with finite planning horizons in stylized example.

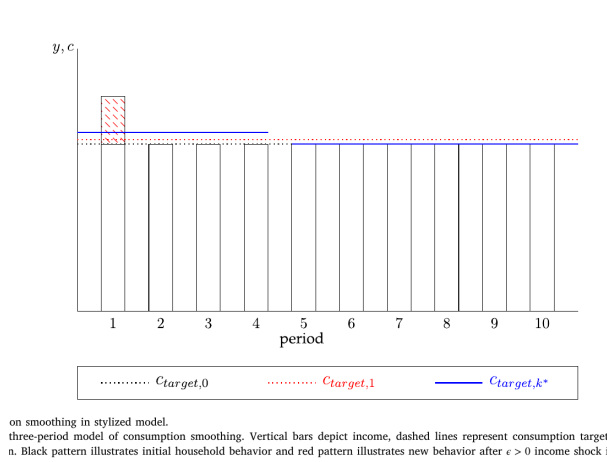
Horizon, k	Cost	MPC in period				
	1	2	3	4	5	
1	$\phi(1)$	1	0	0	0	0
2	$\phi(2)$	$1/2$	$1/2$	0	0	0
3	$\phi(3)$	$1/3$	$1/3$	$1/3$	0	0
4	$\phi(4)$	$1/4$	$1/4$	$1/4$	$1/4$	0

(b) Table 1. MPCs with finite planning horizons.

12.2 Figure 2 and Table 2

Read: Figure 2 extends the smoothing example to a longer horizon. Table 2 reports the long-term model parameters used in the external calibration.

Economic message: The optimal finite horizon is the point at which marginal smoothing benefits no longer justify additional planning costs.



on smoothing in stylized model.
three-period model of consumption smoothing. Vertical bars depict income, dashed lines represent consumption target n. Black pattern illustrates initial household behavior and red pattern illustrates new behavior after $\epsilon > 0$ income shock

(a) Figure 2. Consumption smoothing in longer horizon.

Table 2
Summary of long-term model parameters.

Parameter	Description	Value	Source
Regular income process			
ρ_y	Persistence	0.096	Gelman (2021)
μ_y	Unconditional Mean	0.883	Gelman (2021)
σ_y^2	Variance	0.039	Gelman (2021)
Preferences			
β	Annualized Time Preference	0.941	Kaplan and Violante (2014)
γ	Risk Aversion	4	Kaplan and Violante (2014)
	Elasticity of Intertemporal Substitution	$1/2$	Kaplan and Violante (2014)
Financial markets			
r_a	Annualized Saving Rate	2.73%	Federal Reserve Board
r_d	Annualized Borrowing Rate	9.10%	Survey of Consumer Finances (2007)
$\bar{a}$	Borrowing Limit ($\times$ monthly income)	1.51	Survey of Consumer Finances (2007)

(b) Table 2. Long-term model parameters.

12.3 Figure 3 and Table 3

Read: Figure 3 compares the calibrated model's stationary distribution of liquid wealth to the data. Table 3 summarizes households in the 2008 CEX sample by terciles of stimulus payment relative to monthly income.

Economic message: The calibration targets heterogeneous shock size relative to household resources, which is central for the planning-horizon mechanism.

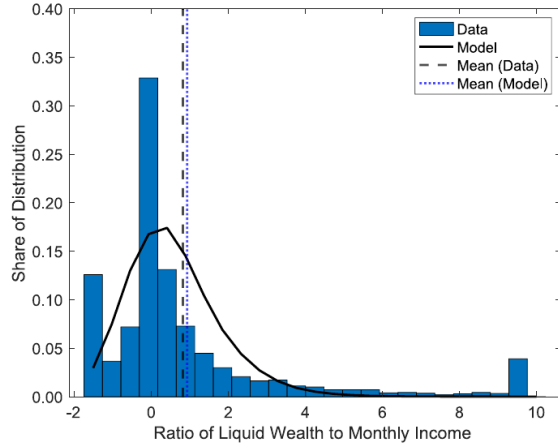


Fig. 3. Stationary distribution of wealth using long-term plans.

Notes: In blue, histogram of liquid wealth to monthly income in the data, censored from below at -1.51 , the borrowing constraint in the model, and from above. Black line shows the stationary distribution of liquid wealth to monthly income in the model. Vertical lines depict the average level of liquid wealth to monthly income in the data and model.

(a) Figure 3. Stationary distribution of wealth.

Table 3
CEX terciles of ESP to monthly inc.

		Mean	Std. Dev.	25th Perc.	Median	75th Perc.
T1	Rel. ESP	0.109	0.036	0.084	0.115	0.138
	ESP Amount	803	413	600	600	1,200
	Monthly Inc.	7,862	3,903	4,681	7,507	10,310
	Liquid Assets	14,127	21,409	1,600	5,788	17,000
T2	Rel. ESP	0.218	0.034	0.189	0.212	0.245
	ESP Amount	1,023	495	600	1,200	1,200
	Monthly Inc.	4,778	2,360	2,833	4,583	6,393
	Liquid Assets	11,750	23,393	500	2,706	10,000
T3	Rel. ESP	1.187	10.302	0.334	0.405	0.560
	ESP Amount	1,048	575	600	1,030	1,200
	Monthly Inc.	2,398	1,536	1,250	2,000	3,388
	Liquid Assets	5,652	15,169	5	900	4,200

(b) Table 3. CEX terciles of ESP to monthly income.

12.4 Table 4 and Table 5

Read: Table 4 reports the spending response to Economic Stimulus Payments. Table 5 reports the GMM targets and external validation moments for the calibrated model.

Economic message: The model is calibrated to match how the MPC varies with shock size relative to income, not only average stimulus spending.

Table 4
Spending response of consumption to economic stimulus payments.

	(a) Non-durables		(b) Durables		(c) Total	
	Estimate	Implied MPC	Estimate	Implied MPC	Estimate	Implied MPC
ESP (Base: Tercile 1)	0.347** (0.168)		0.715 (0.537)		1.062* (0.576)	
ESP $\times$ Tercile 2	-0.137 (0.138)	0.210* (0.120)	-0.081 (0.423)	0.634 (0.392)	-0.217 (0.450)	0.845** (0.404)
ESP $\times$ Tercile 3	-0.232* (0.136)	0.115 (0.109)	-0.260 (0.424)	0.455 (0.347)	-0.492 (0.454)	0.569 (0.368)
Observations	8,592		8,592		8,592	
R ²	0.018		0.005		0.007	

(a) Table 4. Spending response to stimulus payments.

Table 5
Summary of parameter values (external estimates and calibrations).

(a) GMM targets						
#	Description		Target	Model		
1	3M MPC for $\Delta = 0.11y$	(50th Percentile of T1)	0.347	0.336		
2	3M MPC for $\Delta = 0.16y$	(100th Percentile of T1)	0.279	0.277		
3	3M MPC for $\Delta = 0.21y$	(50th Percentile of T2)	0.210	0.207		
4	3M MPC for $\Delta = 0.28y$	(100th Percentile of T2)	0.163	0.156		
5	3M MPC for $\Delta = 0.40y$	(50th Percentile of T3)	0.115	0.115		
6	3M MPC for $\Delta = 0.56y$	(75th Percentile of T3)	0.057	0.103		
(b) External validation						
	$\Delta = 0.33y$		$\Delta = 0.45y$		$\Delta = 0.58y$	
	Data	Model	Data	Model	Data	Model
t = 1	0.083	0.057	0.059	0.042	0.038	0.034
t = 2	0.144	0.113	0.110	0.084	0.075	0.068
t = 3	0.173	0.167	0.138	0.125	0.096	0.101

(b) Table 5. GMM targets and validation.

12.5 Figure 4 and Figure 5

Read: Figure 4 shows the calibrated planning cost function. Figure 5 compares the MPC-liquidity relationship in the data and in alternative models.

Economic message: Modest but increasing planning costs allow the BIR model to generate high MPCs even away from borrowing constraints.

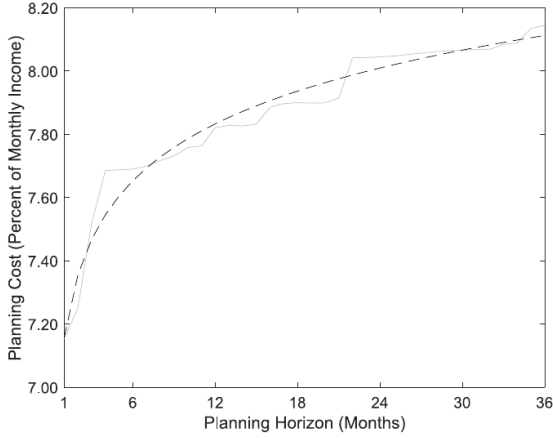


Fig. 4. Calibrated planning costs as fraction of monthly income.
Notes: Gray line is calibrated planning cost as a function of monthly income for household with median wealth and dashed black line is logarithmic approximation.

(a) Figure 4. Calibrated planning costs.

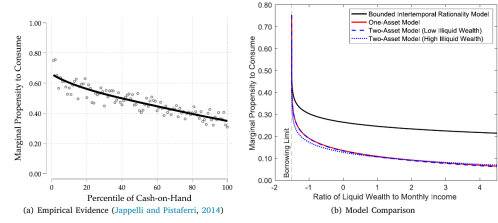


Fig. 5. Marginal propensity to consume and liquid wealth.
Notes: (a) Marginal propensity to consume is survey response to hypothetical question, Cash-on-hand is the sum of income and liquid wealth. Source: 2010 Italian Survey of Household Income and Wealth. Redicates Fig. 7 of Jappelli and Pistaferri (2014). (b) Marginal propensity to consume out of a windfall income shock

(b) Figure 5. MPC and liquid wealth.

12.6 Figure 6 and Figure 7

Read: Figure 6 shows how the fraction of households with a positive MPC rises with shock size. Figure 7 shows how the MPC conditional on responding falls with shock size.

Economic message: The BIR model jointly generates a positive extensive-margin effect and a negative intensive-margin effect.

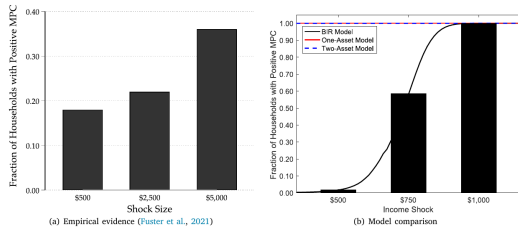


Fig. 6. Fraction of households with positive MPC and shock size.
Notes: (a) Fraction of households with positive MPC in response to hypothetical windfall income shock in survey data. Source: Table 3 in Fuster et al. (2021). (b) Fraction of households with positive MPC in response to income shocks across different models.

(a) Figure 6. Positive MPC and shock size.

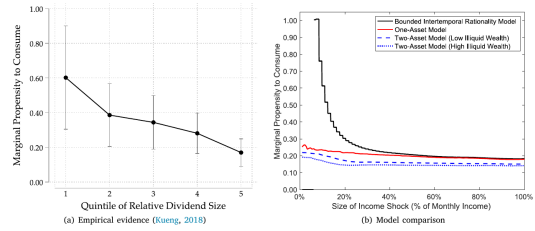


Fig. 7. Marginal propensity to consume and shock size.
Notes: (a) Estimated marginal propensity to consume out of dividend payments from the Alaska Permanent Fund with 95% confidence intervals. Relative dividend size is payment divided by income. Source: Columns (5) and (6) of Table 4 in Kuong (2018). (b) Marginal propensity to consume out of an income shock range from 0% to 100% of monthly income across different models.

(b) Figure 7. MPC and shock size.

12.7 Table 6 and Figure 8

Read: Table 6 summarizes alternative stimulus-program designs and aggregate MPCs. Figure 8 translates reductions in planning costs into equivalent increases in windfall shock size.

Economic message: Stimulus design and financial literacy interventions both work through the endogenous planning horizon, but they affect different parts of the wealth distribution.

Table 6

Summary of fiscal stimulus programs: Payments and aggregate MPCs.

	Benchmark	Alternate 1	Alternate 2
Lowest 25% Income	–	+\$600	+\$600
Middle 50% Income	–	–\$300	–
Highest 25% Income	–	–	–\$600
Aggregate MPC	0.257	0.301	0.275
% Rel. to benchmark	–	17.4%	7.20%

(a) Table 6. Fiscal stimulus programs.

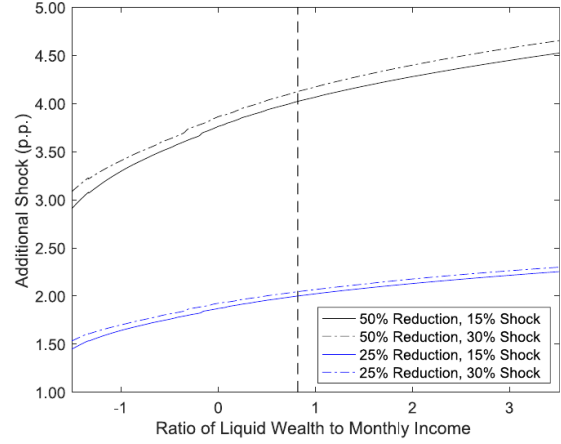


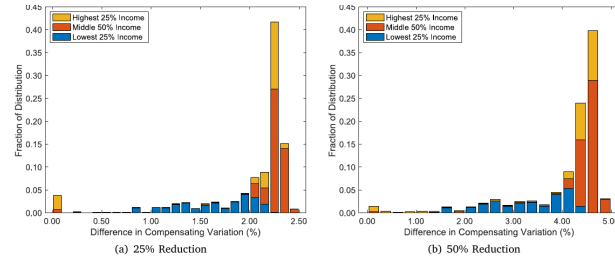
Fig. 8. Comparison of shock sizes with increased financial literacy.
Notes: The additional shock required (in percentage points of monthly income) to make a household indifferent between baseline planning costs and reduced planning costs from increased financial literacy. Dashed vertical line is median

(b) Figure 8. Financial literacy and shock sizes.

12.8 Figure 9

Read: Figure 9 shows the distribution of welfare gains from reductions in planning costs.

Economic message: Lower planning costs can create larger welfare gains for wealthier households because they have more to gain from long-horizon smoothing.



g. 9. Marginal propensity to consume and liquid wealth.
otes: The distribution of welfare gains from a 25% decrease and 50% decrease in planning costs due to increased financial literacy. The distribution of gain is further divided into three groups: the lowest 25% of the income distribution, the middle 50%, and the highest 25%.

Figure 9. Welfare gains from lower planning costs.